National Institute of Technology Patna
Department of Electronics and Communication Engineering
End-Semester Examination – July-Dec 2025

Course: B.Tech-ECE/EE-VLSI/DD-ECE-MV **Semester: I** **Time: 03 Hours**
Code: EC011102/EC011202/EC011302 **Subject: Electronics Workshop** **Full Marks: 60**

Attempt all questions. Assume the missing data, if any.

Q.1 a) Only 10% of the input power appears at the output terminals of a cable. Calculate the attenuation in decibels (dB). **(2)**

b) Suppose you have only the following resistor values available in the lab:

100 Ω, 220 Ω, 330 Ω, 470 Ω, 1 kΩ.

Design the closest possible practical 6 dB symmetrical attenuator (T-type and π-type) for $R_0 = 50\ \Omega$ and compute: (a) Actual attenuation, (b) Mismatch introduced **(8)**

Q.2 a) Consider the commercially available **E12** resistor series. Realize target resistances of 8.87 kΩ and 123.00 Ω using **E12** resistors only. For each target, show 1) the chosen resistor combination, 2) the computed resulting resistance to 2 decimal places, and 3) the percent error. **(6)**

b) Sketch and explain the **I–V characteristics of an LED**. Why does LED require a series/parallel resistor? **(4)**

Q.3 a) What is a PCB? With a neat drawing, show different parts of the PCB, label it, and explain briefly. **(5)**

b) What are the different types of PCBs? Explain each briefly with a neat sketch. **(5)**

Q.4 a) Prepare a PCB layout for a **common-collector (emitter-follower) amplifier** using an NPN transistor. Label input, output, and power terminals. **(4)**

b) Explain the complete procedure using a flowchart for **fabricating a PCB using the wet chemical process.** **(6)**

Q.5 a) Compare CNC milling with chemical etching from an environmental and sustainability perspective. Which is more eco-friendly? **(4)**

b) **Design a conceptual circuit where a photodiode controls the brightness of an LED in a smooth (analog) manner.** You may use **BJT(s), resistors, and diodes only** (no capacitors, regulators, or MOSFETs).

Given / constraints

- Supply: **VCC = 12 V**.
- LED: **V_F = 2.0 V**, desired **I_LED,max = 20 mA**.
- Photodiode photocurrent range (reverse-biased): **I_pd_dark ≈ 0.1 µA**, **I_pd_bright ≈ 80 µA**.
- Limit photodiode reverse bias to **≤ 5 V**.
- LED should be essentially **OFF in dark** (I_LED < 0.5 mA) and reach ≈**20 mA** in bright light.
- Circuit must provide **smooth (continuous)** brightness variation with illumination. **(4)**

c) Explain **cross-talk** in PCB traces. **(2)**

Q.6 a) Why is impedance matching important when designing attenuators?

b) State two practical applications of T-type and pi-type attenuators in electronic circuits.

c) What are the differences between through-holes and vias?

d) What are the different tools used in CNC milling-based PCB fabrication?

e) What is soldering? Why is it needed in making electronic circuits? **(2×5)**

****************************END OF THE QUESTION PAPER****************************